

Şirag Erkol

CONTACT INFORMATION	Northwestern University Center for Science of Science and Innovation 1800 Sherman Ave, Evanston, IL 60201	(812) 558-8938 siragerkol@gmail.com siragerkol.com
ACADEMIC POSITIONS	Northwestern University , Evanston, IL <i>Research Assistant Professor</i>	July 2025 – present
	Northwestern University , Evanston, IL <i>Postdoctoral Fellow</i>	July 2023 – June 2025
	Indiana University , Bloomington, IN <i>Research & Teaching Assistant</i>	August 2017 – May 2023
	Boğaziçi University , Istanbul, Turkey <i>Research & Teaching Assistant</i>	October 2014 – August 2017
EDUCATION	Indiana University , Bloomington, IN <i>Ph.D. in Informatics, Complex Networks and Systems track</i> • Advisor: <i>Assoc. Prof. Filippo Radicchi</i> • Thesis: <i>Identifying Influential Spreaders in Complex Networks</i> • Minor: Computer Science <i>M.S. in Informatics</i>	August 2017 – May 2023
	Boğaziçi University , Istanbul, Turkey <i>M.S. in Industrial Engineering</i> • Advisor: <i>Assoc. Prof. Gönenç Yücel</i> • Thesis: <i>Influence Maximization Based on Partial Network Structure Information: A Comparative Analysis on Seed Selection Heuristics</i> <i>B.S. in Industrial Engineering</i>	September 2014 – August 2016 September 2010 – June 2014
	• Graduation Project: <i>Analyzing Parking Stand Requirements for Istanbul Ataturk Airport</i>	
WORK EXPERIENCE	Pacific Northwest National Laboratory , Richland, WA <i>Ph.D. Intern</i>	June 2021 – August 2021
HONORS & AWARDS	IU Luddy Summer Research Award (\$4400)	2022
PUBLICATIONS	Published: Crowding controls the scaling of bus frequency with demand S. Patwardhan, Ş. Erkol , F. Radicchi, M. Barthelemy, <i>PNAS</i> , 123, e2535998123 (2026) [DOI] [arXiv] Symmetry breaking in optimal transport networks S. Patwardhan, M. Barthelemy, Ş. Erkol , S. Fortunato, F. Radicchi, <i>Nature Communications</i> , 15, 3758 (2024) [DOI] [arXiv] Consistency pays off in science Ş. Erkol , S. Sikdar, F. Radicchi, and S. Fortunato, <i>Quantitative Science Studies</i> , 4, 491 (2023) [DOI] [arXiv] [GitHub] [Website]	

Effective submodularity of influence maximization on temporal networks
Ş. Erkol, D. Mazzilli and F. Radicchi, *Physical Review E*, 106, 034301 (2022)
[\[DOI\]](#) [\[arXiv\]](#)

Who is the best coach of all time? A network-based assessment of the career performance of professional sports coaches
Ş. Erkol and F. Radicchi, *Journal of Complex Networks*, 9, cnab012 (2021)
[\[DOI\]](#) [\[arXiv\]](#) [\[Website\]](#)

Influence maximization on temporal networks
Ş. Erkol, D. Mazzilli and F. Radicchi, *Physical Review E*, 102, 042307 (2020)
[\[DOI\]](#) [\[arXiv\]](#)

Systematic comparison between methods for the detection of influential spreaders in complex networks
Ş. Erkol, C. Castellano and F. Radicchi, *Scientific Reports*, 9, 15095 (2019)
[\[DOI\]](#) [\[arXiv\]](#)

Influence maximization in noisy networks
Ş. Erkol, A. Fageeh and F. Radicchi, *EPL*, 123, 58007 (2018)
[\[DOI\]](#) [\[arXiv\]](#)

Influence Maximization Based on Partial Network Structure Information: A Comparative Analysis on Seed Selection Heuristics
Ş. Erkol and G. Yücel, *International Journal of Modern Physics C*, 28, 1750122 (2017)
[\[DOI\]](#)

PRESENTATIONS

Talks:

Second acts in scientific and creative careers **June 2026**
 NetSci 2026, Boston, MA, USA

Effective submodularity of influence maximization on temporal networks **July 2022**
 NetSci 2022, Online

Influence maximization on temporal networks **July 2021**
 Networks2021, Online

Influence maximization on temporal networks **March 2021**
 University of Limerick, Ireland, Online

Posters:

Second acts in scientific and creative careers **June 2026**
 ICSSI 2026, Boulder, CO, USA

Consistency pays off in science **June 2023**
 ICSSI 2023, Evanston, IL, USA

Systematic comparison between methods for the detection of influential spreaders in complex networks **September 2020**
 NetSci 2020, Online

Influence maximization in noisy networks **May 2019**
 NetSci 2019, Burlington, VT, USA

COMMUNITY SERVICE

Journal Reviewer: Applied Mathematics and Computation; Chaos, Solitons & Fractals; EPJ Data Science; Expert Systems with Applications; IEEE Transactions on Network Science and Engineering; Journal of Complex Networks; Journal of Computational Science; Pattern Recognition; Physical Review E; Royal Society Open Science; Science Advances

Conference Reviewer: ISSI (2023); ICSSI (2024); IC2S2 (2026)

Journal Subreviewer: PLoS One, PNAS

COMPUTER
SKILLS

Frequent user of Python, $\text{\LaTeX}$
Experience in Django, NetLogo, HTML, CSS
Familiar with MySQL, C, R, Git, Stella

LANGUAGES

English (fluent), Turkish (native), Armenian (native), French (beginner), Spanish (beginner)

Last updated: October 2026